FACULTY OF COMPUTING
UTM Johor Bahru

SECV2223-04 WEB PROGRAMMING

Semester 02, 2025/2026

Assignment 1

Lecturer:

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19th April 2026

1. Websites Review

1.1 Universiti Teknologi Malaysia (UTM) Official Website

Link: [UTM's Official Website]

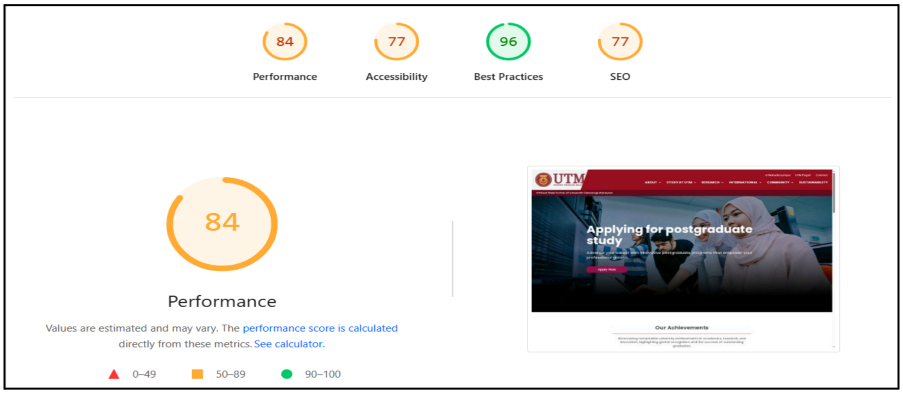

Figure 1: Performance for UTM's Official Website

Referring to Figure 1, the performance report shows that the website performs well with a score of 84, supported by low Total Blocking Time (10 ms) and minimal Cumulative Layout Shift (0.036), indicating smooth interactivity and stability. However, weaknesses include a large page size (6 MB), unused CSS (621 KB) and JavaScript (397 KB), and render-blocking resources. Accessibility and SEO scores of 77 also reveal issues such as poor contrast, improper heading structure, missing semantic elements, and lack of meta description and robots.txt. The evaluation was based on core metrics and diagnostic insights. Some improvements the website could use would include optimizing images, reducing unused code, deferring scripts, improving HTML structure, enhancing contrast, and implementing proper SEO elements to improve overall performance and usability.

1.2 Harvard University Official Website

Link: [Harvard University Official Website]

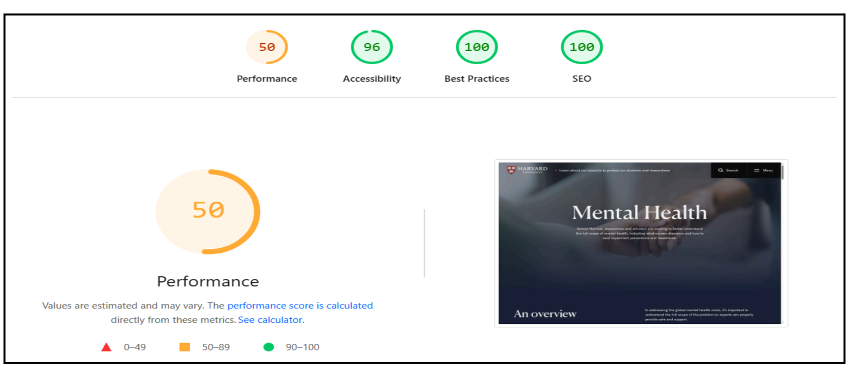

Figure 2: Performance Report on Harvard University's Website

Referring to Figure 2, the performance report shows that the website has strong strengths in accessibility, best practices, and SEO, with scores of 96, 100, and 100, indicating good structure and optimization. However, its performance score of 50 highlights weaknesses, mainly due to high Total Blocking Time (2,720 ms), heavy JavaScript execution, and large image sizes affecting loading speed. The evaluation was conducted by analyzing key metrics such as FCP, LCP, TBT, and diagnostic insights. Based on these findings, improvements for the website should focus on optimizing images, reducing and deferring JavaScript, minimizing unused code, and enhancing caching to improve overall performance and responsiveness.

1.3 Massachusetts Institute of Technology

Link: https://www.mit.edu/

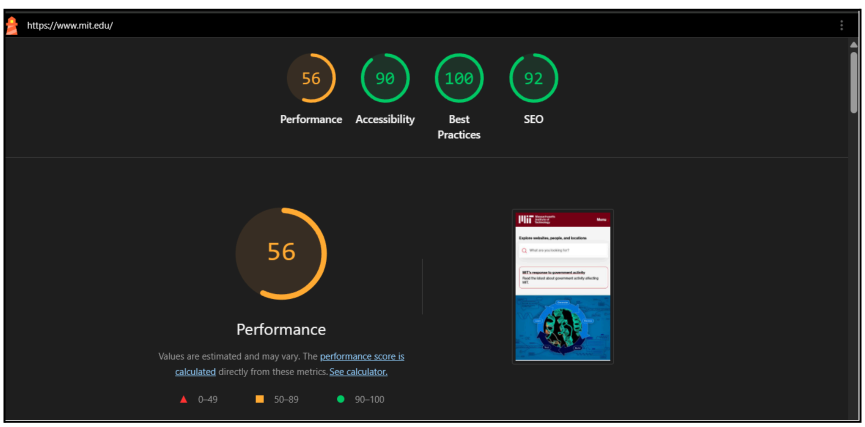

Figure 3: Performance Report on Massachusetts Institute of Technology's Website

Referring to Figure 3, the performance report shows that the website has strong strengths in accessibility, best practices, and SEO, with scores of 90, 100, and 92, indicating good structure and optimization. However, its performance score of 56 highlights weaknesses, mainly due to modern HTTP, Render Blocking Request, Use efficient cache lifetimes and Font display affecting loading speed. The evaluation was conducted by analyzing key metrics LCP, Layout shift culprits, Network dependency tree, cache lifetimes and render-blocking. Based on these findings, improvements for the website should focus on improving image delivery by optimizing image formats, compressing file sizes, and implementing lazy loading to enhance loading speed and overall performance.

1.4 Universiti Malaya (UM)

Link: University Malaya Website

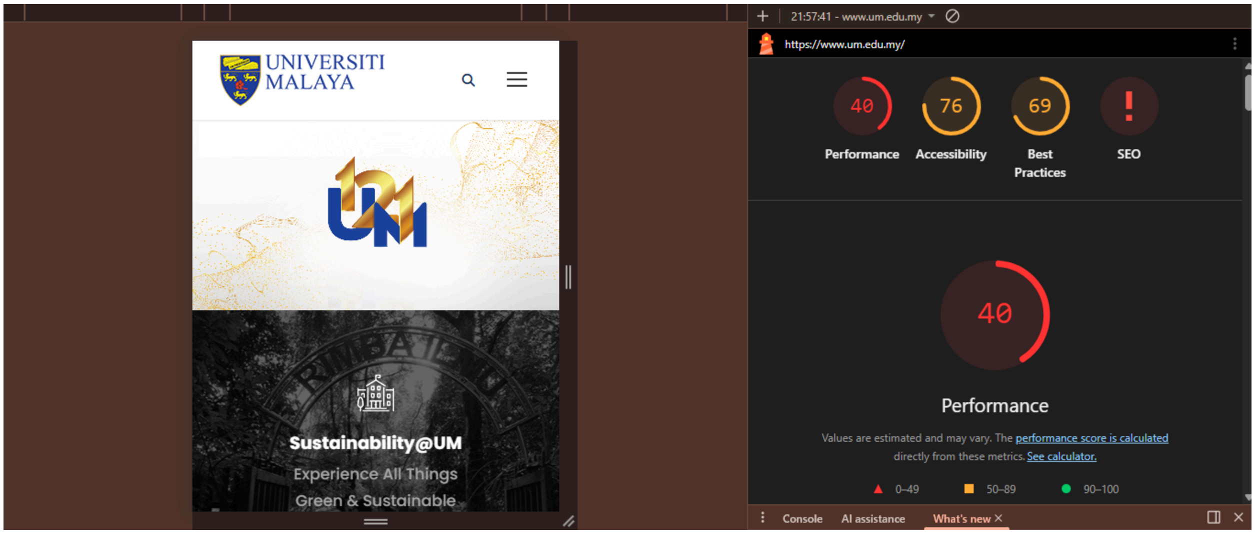

Figure 5: Performance Report on University Malaya's Website

Referring to Figure 5, the performance shows a low performance score of 40, which means the website may load slowly and affect user experience. The accessibility score is 76, indicating it is fairly usable but still needs improvement for better support of all users. The best practices score is 69, showing that some web standards are followed but there are still issues to fix. The SEO score is low, suggesting the website is not fully optimized for search engines. Overall, the website works but requires improvements in speed, accessibility, and optimization, such as reducing large file sizes, fixing accessibility issues, and improving SEO elements to provide a smoother and more user-friendly experience.

1.5 Universiti Kebangsaan Malaysia (UKM)

Link:[Universiti Kebangsaan Malaysia Website]

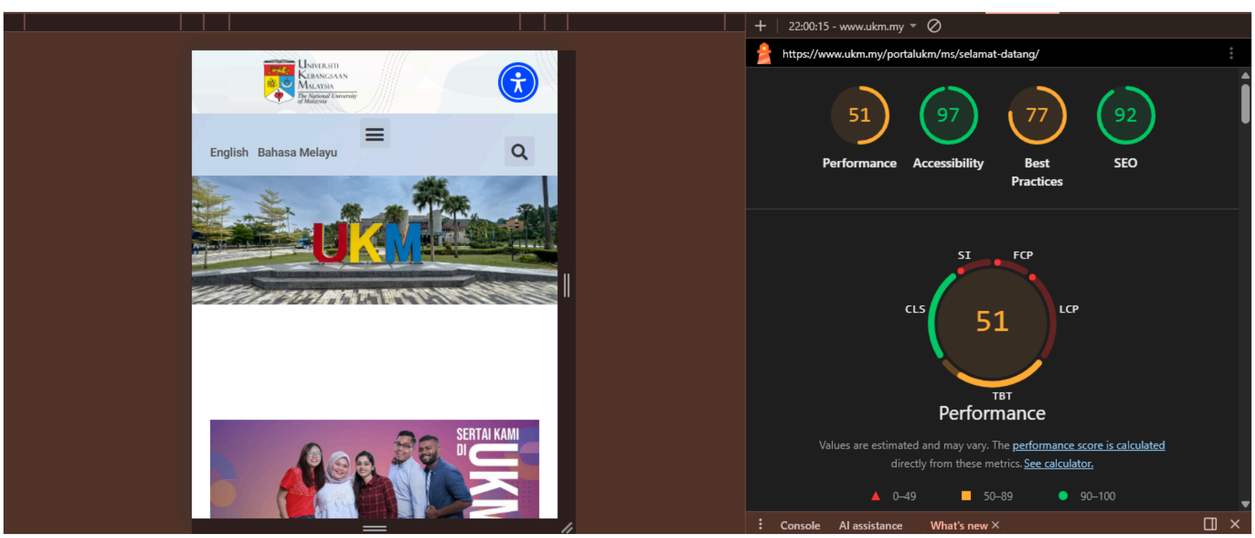

Figure 6: Performance Report on University Malaya's Website

Referring to Figure 6, the performance shows a moderate performance score of 51, meaning the website speed is acceptable but still can be improved. The accessibility score is very high at 97, indicating the website is well-designed for users, including those with disabilities. The best practices score is 77, showing that most web standards are followed but there are still some areas to improve. The SEO score is also high at 92, meaning the website is well optimized for search engines. Overall, the website performs well, especially in accessibility and SEO, but can still improve in performance and best practices.

2.0 Comparative Evaluation and Trend Analysis

The evaluation of the five selected university websites UTM, Harvard, MIT,UM, and UKM reveals a significant disparity between performance and accessibility across the higher education domain. While the international institutions like Harvard and MIT were selected based on QS World University Rankings, the local institutions provide a baseline for regional performance.

- **Performance vs Content Density:** A notable trend is that websites with highest Best Practices and SEO scores like Harvard at 100 and MIT at 92 suffer from the lowest Performance score like 50 and 56 respectively. This is primarily due to heavy javascript execution and high Total Blocking Time (TBT).
- **Regional Leader:** UTM stands out as the performance leader with a score 64, indicating a more streamlined front end architecture compared to others.
- **Accessibility Standards:** With the exception of UM (76) and UTM (77), most sites achieved scores above 90, demonstrating that modern academic institutions prioritize inclusive web design.

3.0 Strategic Recommendations or Optimization

To improve the overall user experience and technical efficiency of these domains, the following optimizations are recommended:

- **Resource Management:** Implement code splitting and lazy loading for javascript to reduce the high Total Blocking observed in the Harvard and MIT audits.
- **Media Optimization:** Convert all high resolution imagery to modern formats like WebP and utilize efficient cache lifetime to resolve the slow loading speeds identified for UM and UKM.
- **SEO Foundation:** For UTM and UM, it is critical to implement missing meta descriptions and robot.tst files to improve search engine crawling and visibility.
- **Structural Integrity:** Address contrast issues and improper heading structures like H1-H6 to bring all institutions up to the Excellent accessibility tier.